Efficient Low-Rank Representation for Hyperspectral Anomaly Detection via Pixel Segmentation

Kun Yu, Zebin Wu, *Senior Member, IEEE*, Jin Sun, *Member, IEEE*, Yang Xu, *Member, IEEE*, Yi Zhang, Zhihui Wei, *Member, IEEE*, and Peng Zheng

***Abstract*— Low-rank representation is a popularly used method in remote sensing and image processing systems. Existing low-rank representation-based algorithms often neglect the varying sensitivities of different pixels to the dictionary, which may lead to inaccurate detection. Also, the dense iterative computations involved in these algorithms could induce high computation overheads. This article proposes an efficient low-rank representation method for hyperspectral anomaly detection (HAD) on a central processing unit (CPU)-field programmable gate array (FPGA) hybrid computing platform. The proposed method starts with segmenting the hyperspectral image into superpixels based on the selected number of dictionary categories. The image is divided into center and edge parts using a sliding window along the segmentation boundaries. Then, we use a low-rank and sparse representation (LRASR) to process center pixels that are well-adapted to the dictionary, and a weighted low-rank and collaborative representation model to process edge pixels that are less adaptive to the dictionary. The aforementioned two representation models are integrated to obtain the final reconstructed image. Moreover, we propose the Nesterov acceleration method by incorporating an adaptive step size into the linearized alternating direction method with adaptive penalty (LADMAP). At the hardware level, the acceleration method is deployed on the CPU-FPGA platform to speed up the proposed HAD flow. Experimental results demonstrate that the proposed method outperforms most existing methods in detection accuracy and achieves significant improvement in computational efficiency.**

***Index Terms*— central processing unit (CPU)-field programmable gate array (FPGA), hardware acceleration, hyperspectral anomaly detection (HAD), low-rank representation, Nesterov.**

I. Introduction

HYPERSPECTRAL images (HSIs) are captured by specialized hyperspectral imaging sensors and consist of hundreds of spectral bands, containing rich spatial and spectral information. Anomaly detection identifies pixels or regions that deviate from the statistical characteristics of an image. These anomalies often exhibit spectral features significantly different from their surroundings, potentially indicating abnormal materials, rare targets, or valuable features. Currently, hyperspectral anomaly detection (HAD) has been widely applied in areas such as environmental monitoring, agriculture, and military applications [1], [2], [3].

In order to accurately obtain the anomalies in the image, numerous research methods have emerged in the field of HAD. Among them, statistical detection methods and representation-based detection methods are two major strategies. The Reed–Xiaoli (RX) algorithm, proposed in 1990 [4], is the most widely used statistical model and serves as a benchmark for HAD. The RX algorithm assumes that the background pixels in an HSI follow a multivariate Gaussian distribution. For each pixel, it calculates the Mahalanobis distance from the background mean to assess the deviation of the pixel from the background. While the RX algorithm is computationally efficient and conceptually straightforward, its performance is limited in scenarios with complex background structures or high noise levels. To overcome these limitations, subsequent researchers have proposed many improved versions of the RX algorithm. For example, the local RX (LRX) algorithm [5] and the kernel RX (KRX) [6] algorithm were proposed, which are more sensitive to anomalies and offer better detection performance. These advancements aim to more effectively isolate anomalous features from HSI, addressing the inherent challenges of diverse spectral characteristics and complex backgrounds.

In addition to the popular RX algorithm and its variants, representation-based methods also play an important role in HAD. Unlike traditional statistical methods, this category of methods does not rely on the strong assumptions typically required by statistical techniques. Instead, they leverage properties such as sparsity, low-rankness, or dictionary learning to identify anomalies. Each pixel in an HSI can be represented as a linear combination of other pixels or a dictionary. If a pixel is difficult to represent in this manner, it may be identified as an anomaly. Moreover, compared with deep methods [7], [8], which have been popularly used in HAD, representation-based methods are advantageous in efficiency and interpretability. While deep neural networks (DNNs) [9], [10], [11], [12] and convolutional neural networks (CNNs) [13], [14] demonstrate their efficacy in anomaly detection by abstracting features to

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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: 319106010139@njust.edu.cn; wuzb@njust.edu.cn; sunj@njust.edu.cn; xuyangth90@njust.edu.cn; yzhang@njust.edu.cn; gswei@njust.edu.cn; penn0121@163.com).

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separate anomalies from the background, they often suffer from complex computations and unpredictable behaviors.

To date, a multitude of representation-based algorithms for HAD has been developed. Based on the characteristics of data structures, representation-based methods can be categorized into low-rank representation, sparse representation, and collaborative representation, which can be combined to adapt to different types of hyperspectral data. The low-rank and sparse representation (LRASR) method proposed by Xu et al. [15] utilizes the low-rank and sparse characteristics of the coefficient matrix to capture the essential features of the data and reconstruct the background matrix. Li et al. [16] proposed an improved algorithm based on LRASR and joint Gaussian mixture distribution (MOG-LRASR), modeling the sparse components as a Gaussian mixture distribution to obtain a cleaner low-rank background component. Moreover, representation-based methods often require learning and updating dictionary atoms to achieve better results. Zhang et al. [17] proposed an update method for spectral difference low-rank dictionary representation learning, which learns differences between adjacent bands. The integration of tensor modeling into low-rank representation has emerged as a new research trend [18], [19], [20]. By leveraging tensor-based low-rank modeling, anomalies can be more effectively highlighted. Recent studies have further advanced this approach by incorporating deep learning with low-rank representation, leading to more powerful detection frameworks. Li et al. [21] proposed a low-rank neural network, which synergizes the low-rank representation model with deep learning techniques. Lin et al. [22] introduced a method that combines total variation regularization with low-rank modeling to enhance robustness.

While representation-based algorithms perform well in hyperspectral data processing tasks, their detection results heavily depend on good dictionary construction. Existing works have focused on dictionary learning but often overlook the varying sensitivities of different pixels to the dictionary. Moreover, these methods often involve matrix decomposition operations, such as singular value decomposition (SVD), which have a high computational complexity when processing large-scale hyperspectral data, resulting in long computation times and high resource consumption. Solving these models also requires iterative optimization techniques, which may require numerous iterations to converge, further increasing computational costs.

To address the aforementioned issues, this article proposes an efficient low-rank representation based on pixel segmentation (ELRBPS) for enhanced hyperspectral data representation and analysis. Specifically, based on the selected number of dictionary categories, superpixel segmentation is used to divide the HSI into several superpixel regions with similar spectral characteristics. Then, a sliding window technique is employed, dividing the image into two parts along the superpixel segmentation lines: a central part and an edge part. Pixels in the central part, due to their high adaptability to the dictionary, are represented using a low-rank and sparse model, which can effectively capture the low-rank structure and sparse features in the data, leading to accurate image reconstruction. For edge pixels, which have poor dictionary adaptability and more complex spectral characteristics, a weighted low-rank and collaborative representation detector (LRCRD) is introduced to better handle these intricate spectral features. After obtaining the representation results for both the central and edge parts, these two sections are integrated to produce the final reconstructed image. This dual-model combination effectively enhances the quality of the reconstructed image, particularly excelling in regions with complex spectral characteristics. In addition, the traditional model-solving process has been optimized in this article. In solving the LRASR problem, this article employs the linearized alternating direction method with adaptive penalty (LADMAP) [23] and further incorporates the Nesterov algorithm with adaptive step size into LADMAP (ALADMAP). Nesterov improves convergence speed during gradient descent by utilizing predictive steps, thereby reducing the number of iterations and significantly speeding up the model-solving process. For efficient image representation and analysis, the optimized algorithm is deployed on a central processing unit (CPU)-field programmable gate array (FPGA) hybrid computing platform to leverage hardware acceleration. The CPU handles flexible tasks such as hyperspectral data preprocessing and dictionary construction, enabling efficient preparation of input data for analysis. Meanwhile, the FPGA is utilized to execute computationally intensive iterative processes, offering high-throughput parallelism and significantly boosting computational performance. The main contributions of this article can be summarized as follows.

1) We propose a novel low-rank representation framework that initially segments pixels based on their similarity to a dictionary and subsequently represents these segmented pixels using two distinct low-rank models.
2) We propose a similarity weight to LRCRD, allowing pixels to be represented by more similar dictionary atoms.
3) We propose ALADMAP for solving the low-rank model, significantly reducing the number of iterations required for convergence.
4) We deploy the algorithm on a CPU-FPGA hybrid computing platform, leveraging parallel optimization techniques to greatly accelerate the processing speed of the data.

The remainder of this article is organized as follows. Section II introduces the related work on HAD. Section III provides a detailed introduction to the proposed low-rank model. Section IV presents the optimization of the LADMAP and parallel process. Section V demonstrates the reliability and acceleration effect of the method through experimental evidence. Finally, Section VI concludes this article.

II. Related Work

A. HAD Based on Low-Rank Representation

Low-rank representation refers to decomposing a data matrix under the constraint of low-rankness, meaning that the original data can be approximated using a smaller set of basis vectors. Hyperspectral data typically exhibit strong spectral correlation, as spectral curves within the same

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scene often share similar patterns. As a result, the original high-dimensional data can often be approximated using a low-rank matrix, simplifying data representation and processing. In recent years, HAD methods based on low-rank representation have gained widespread attention. For instance, Yang et al. [24] proposed a new abundance and dictionary-based low-rank representation (ADLR) method for anomaly detection. Chang et al. [25] used orthogonal subspace projection (OSP) to find low-rank and sparse matrices, enhancing the representation ability and detection accuracy. Zhang et al. [17] proposed a dictionary learning method based on the relationship between adjacent bands, which can learn a dictionary that reconstructs the background but not the anomalies. In addition to research that transforms hyperspectral data into matrices and imposes low-rank constraints, studies on low-rank tensor representation have also rapidly developed. Sun et al. [20] proposed an adaptive low-rank transformation-based HAD algorithm, which decomposes the input HSI into a background tensor, anomaly tensor, and noise tensor. In addition, Qin et al. [26] introduced the Generalized nonconvex low-rank tensor representation (GNLTR) for HAD, providing a unified solver capable of addressing the challenges in existing nonconvex optimization problems.

B. Optimization of HAD

Due to its high dimensionality, strong spectral correlation, and complex characteristics, hyperspectral data pose significant challenges for traditional detection algorithms, including high computational costs, limited real-time performance, and sensitivity to noise. To address these challenges, optimization strategies focus on algorithm improvement, computational acceleration, and model enhancement. Xu et al. [27] proposed an anomaly detection algorithm on cloud platforms from the perspective of data filtering; in their work, they use a scoring mechanism to exclude data with minimal impact on training results, retaining data with higher influence factors. Molero et al. [28] proposed a multicore processing RX algorithm and its variants, which improve performance while maintaining accuracy. Du et al. [29] analyzed the RX algorithm, discussing its real-time processing version, GPU implementation, and cloud and FPGA implementations. Macias et al. [30] also proposed an FPGA hardware-optimized implementation of the automatic target detection and classification algorithm (ATDCA) based on the Gram–Schmidt (GS) method, which outperforms implementations on multicore processors and GPU acceleration. Tu et al. [31] proposed an HAD method based on frequency-domain analysis, enhancing anomaly detection through low-pass filtered amplitude spectra and phase reconstruction. By integrating quaternion Fourier transforms for multiscale and multifeature analysis, this method significantly enhances both detection accuracy and computational efficiency [31].

III. Algorithm

In this section, we introduce the principles behind the algorithm. First, we introduce LRASR and LRCRD. Then, we present the framework of our algorithm.

A. Low-Rank Representation

Hyperspectral data are a 3-D image that can be transformed into a matrix $\boldsymbol{X} \in \mathbb{R}^{N\times L}$ during processing, where N represents the number of pixels and L denotes the number of spectral bands in the image. In HSIs, anomalous pixels typically exhibit significant differences from background pixels, and background pixels often exhibit strong correlations, meaning that a background pixel can be represented as a linear combination of other background pixels. This characteristic suggests that the hyperspectral matrix can be decomposed into a background pixel component and an anomalous pixel component as

$$\boldsymbol{X} = \boldsymbol{DS} + \boldsymbol{E} \tag{1}$$

where $\boldsymbol{DS}$ represents the background of the image, $\boldsymbol{D} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \ldots, \boldsymbol{d}_m]$ represents dictionary of background pixels, $\boldsymbol{S}$ represents the coefficient matrix of all pixels, and $\boldsymbol{E}$ denotes the residual matrix, which physically represents the anomalous portion of the image. Since anomalous pixels occupy only a small portion of the entire image, the matrix $\boldsymbol{E}$ can be considered column-sparse. On the other hand, each pixel corresponds to either a single material (pure pixel) or a mixture of multiple substances (mixed pixel), with the spectral curves of each material representable within a certain subspace. Thus, all pixels in HSI can be viewed as distributed across multiple subspaces, and the coefficient matrix $\boldsymbol{S}$ should represent the low-rank approximation of all pixels.

1) LRASR: In anomaly detection, since most pixels belong to background samples, these samples can be sparsely represented using a background dictionary. Therefore, the sparsity property of matrix $\boldsymbol{S}$ can effectively describe the local spectral characteristics of all pixels, and the low-rank representation model with a sparse regularization term can be expressed as follows:

$$\begin{aligned} &\min \ \|\boldsymbol{S}\|_* + \beta\|\boldsymbol{S}\|_1 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{DS} + \boldsymbol{E} \end{aligned} \tag{2}$$

where $\|\boldsymbol{S}\|_*$ denotes the nuclear norm of the matrix, used to capture the low-rank characteristics, and $\|\boldsymbol{S}\|_1$ denotes the l_1 norm of the matrix, which is the sum of the absolute values of all its elements. The parameters β and λ are used to balance the weights between the low-rank term and the sparse term. Equation (2) incorporates both the global correlations among all pixels in the HSI and the local spectral preservation characteristics of individual pixels, leading to a more accurate description of the original data. Therefore, this model can represent the anomalous pixels in the residual matrix $\boldsymbol{E}$ more accurately.

2) LRCRD: When the dictionary is incomplete, solely using sparsity constraints may not be sufficient to accurately represent the samples, and leveraging collaboration among samples can more effectively capture the structural features of the data. Therefore, replacing the traditional l_1 norm with the l_2 norm to constrain the coefficient matrix $\boldsymbol{S}$ allows the model to better utilize the collaboration among samples, thereby enhancing the representation effectiveness. The specific optimization model

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is given as follows:

$$\begin{aligned} &\min \ \|\boldsymbol{S}\|_* + \beta|\boldsymbol{S}\|_2 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{D}\boldsymbol{S} + \boldsymbol{E} \end{aligned} \tag{3}$$

where $\|\boldsymbol{S}\|_2$ promotes the collaborative participation of all atoms in the dictionary, leading to a more comprehensive background representation while keeping each coefficient small, effectively avoiding information loss due to over-sparsity. $\|\boldsymbol{E}\|_{2,1}$ imposes constraints on the residual matrix to capture the local features of anomalous pixels, thereby depicting anomalies more accurately. This optimization model can comprehensively utilize global and local information, enhancing the ability to distinguish between the background and anomalies in HSIs.

Both models aim to decompose the hyperspectral data into a background component and an anomalous component. However, they differ in their approach to constraining the coefficient matrix $\boldsymbol{S}$, which represents the weights or coefficients of the background dictionary in reconstructing the image. LRASR leverages sparsity constraints to identify anomalies, while LRCRD emphasizes collaboration among samples to improve the overall representation of the background. Both models have their unique strengths and are designed to address different challenges in HAD.

B. ELRBPS

In HAD, LRASR depends on high-quality dictionary construction. To enhance the representativeness and effectiveness of the dictionary, Xu et al. [15] proposed a k-means clustering-based dictionary construction strategy in their papers. This method first clusters all pixels in the HSI and then applies the RX operator to the pixels in each cluster to identify those most likely belonging to the background, from which a subset is selected to form the dictionary.

Unlike methods that randomly select pixels, this clustering-based strategy effectively avoids the dictionary containing only a few types of dominant background material pixels, thereby improving the adaptability of the dictionary and the diversity in complex scenes. However, in HSIs, due to the complexity of the spectra, most pixels are mixed pixels, while the number of pure pixels is often only a few dozen to a few hundred. Therefore, most atoms in the dictionary are also mixed pixels, and there are significant compositional differences between them and other mixed pixels to be tested, making it difficult to accurately represent the test samples through simple linear combinations. Within each clustering category, dictionary atoms are usually pixels with a high degree of spectral homogeneity because they have smaller Mahalanobis distances within the entire category and are closer to the cluster center. This construction method makes it easier for dictionary atoms to represent similar pixels, but, for those pixels that are far from the dictionary atoms, the representation capability of the model is relatively weak, leading to unsatisfactory accuracy.

To address the limitations of poor representation, this article proposes a strategy that divides pixels into central points and edge points. Using the superpixel linear clustering segmentation (SLCS) algorithm, the HSI is partitioned into multiple superpixel regions, as shown in Fig. 1. In each superpixel block, pixels located at the center are defined as central points because they have a smaller spectral distance to the dictionary atoms, resulting in better representation.

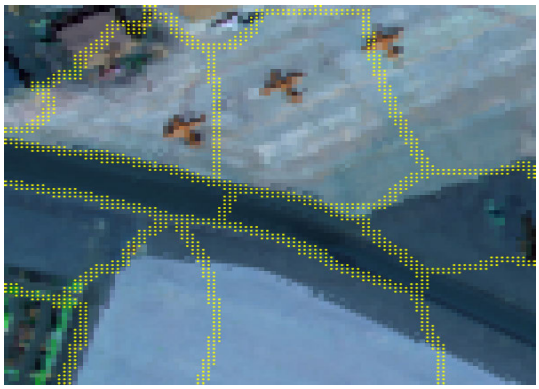


Fig. 1. HSI spilt result.

Pixels distributed around the superpixel boundaries are considered edge points, as their spectral characteristics are more complex, making them less adaptive to the dictionary. The width of the edge pixels is determined by a sliding window applied along the segmentation boundaries. Specifically, the window size controls how many pixels are classified as edge pixels. A larger window results in a wider edge region, while a smaller window leads to a narrower transition area. This division method enables us to design more suitable representation models for pixels in different regions, thereby improving the overall accuracy and efficiency of anomaly detection.

We conducted tests on hyperspectral datasets, employing the LRASR model to detect pixels within different regions of the HSI. The results revealed that the complete image achieved a detection accuracy of 90.03%, whereas the detection accuracy for central points, utilizing the same methodology, reached 94.81%. Conversely, the edge points demonstrated a detection accuracy of only 88.06%. These findings suggest that hyperspectral pixels exhibit significant variability in their sensitivity to the dictionary, which can be attributed to differences in the similarity between the pixels and the dictionary atoms.

In this article, we utilize the LRASR to detect anomalies in central points. For edge points, we propose a spectral similarity weight $\boldsymbol{W}$ based on the LRCRD, which represents the similarity between the dictionary atoms and the spectra being tested

$$\boldsymbol{W}(i,j) = \frac{\boldsymbol{dot}(\boldsymbol{A}(:,i), \boldsymbol{X}(:,j))}{\|\boldsymbol{A}(:,i)\| * \|\boldsymbol{X}(:,j)\|} \tag{4}$$

where $\boldsymbol{dot}(\boldsymbol{A}(:,i), \boldsymbol{X}(:,j))$ computes the inner product of the ith column of $\boldsymbol{A}$ and the jth column of $\boldsymbol{X}$, and $\|\boldsymbol{A}(:,i)\|$ and $\|\boldsymbol{X}(:,j)\|$ are the norms of these two vectors. Through this constraint, we aim for pixels to be represented by dictionary atoms that are more similar, and the model for edge pixels is

$$\begin{aligned} &\min \ \|\boldsymbol{S}\|_* + \beta|\boldsymbol{W}\boldsymbol{S}\|_2 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{D}\boldsymbol{S} + \boldsymbol{E}. \end{aligned} \tag{5}$$

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IV. Optimization

A. LADMAP With Adaptive Step Size Nesterov

In this section, we will elaborate on the solution process of the proposed model and discuss the parallelization and optimization of the employed algorithms to achieve efficient model-solving. Since the solution methods for the LRCRD are nearly identical to those for the LRASR, we will focus solely on the solution and optimization process of the LRASR in this section. By employing the LADMAP, the LRASR model can be solved rapidly. This algorithm linearizes the quadratic penalty term in the augmented Lagrangian method (ADM) and introduces a proximal term when solving subproblems, thereby simplifying the subproblem-solving process. Moreover, LADMAP introduces an adaptive penalty parameter update rule to accelerate convergence. Compared to traditional ADM methods, LADMAP avoids the introduction of auxiliary variables and cumbersome matrix inversion operations, reduces the complexity of the algorithm, and has proven global convergence. When applied to low-rank representation problems, LADMAP integrates sparse SVD techniques to reduce the computational complexity from $O(n^3)$ to $O(rn^2)$, making it suitable for large-scale applications.

For the LRASR, we first introduce an auxiliary variable $\boldsymbol{U}$, letting $\boldsymbol{S} = \boldsymbol{U}$ so that the problem becomes

$$\begin{aligned} &\min \ \|\boldsymbol{S}\|_* + \beta\|\boldsymbol{U}\|_1 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{D}\boldsymbol{S} + \boldsymbol{E}, \quad \boldsymbol{S} = \boldsymbol{U}. \end{aligned} \tag{6}$$

The Lagrangian equation is

$$\begin{aligned} &\boldsymbol{L}(\boldsymbol{S}, \boldsymbol{U}, \boldsymbol{E}, \boldsymbol{Y}_1, \boldsymbol{Y}_2, \mu) \\ &= \|\boldsymbol{S}\|_* + \beta\|\boldsymbol{U}\|_1 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\quad + \langle \boldsymbol{Y}_1, \boldsymbol{X} - \boldsymbol{D}\boldsymbol{S} - \boldsymbol{E}\rangle + \langle \boldsymbol{Y}_2, \boldsymbol{S} - \boldsymbol{U}\rangle \\ &\quad + \frac{\mu}{2}\left(\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S} - \boldsymbol{E}\|_F^2 + \|\boldsymbol{S} - \boldsymbol{U}\|_F^2\right) \end{aligned} \tag{7}$$

$$\begin{aligned} &= \|\boldsymbol{S}\|_* + \beta\|\boldsymbol{U}\|_1 + \lambda\|\boldsymbol{E}\|_{2,1} \\ &\quad + f(\boldsymbol{S}, \boldsymbol{U}, \boldsymbol{E}, \boldsymbol{Y}_1, \boldsymbol{Y}_2, \mu) - \frac{1}{2\mu}\left(\|\boldsymbol{Y}_1\|_F^2 + \|\boldsymbol{Y}_2\|_F^2\right) \end{aligned} \tag{8}$$

where $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ are Lagrange multipliers, μ is the penalty parameter, and

$$\begin{aligned} &f(\boldsymbol{S}, \boldsymbol{U}, \boldsymbol{E}, \boldsymbol{Y}_1, \boldsymbol{Y}_2, \mu) \\ &= \frac{\mu}{2}\left(\left\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S} - \boldsymbol{E} + \frac{\boldsymbol{Y}_1}{\mu}\right\|_F^2 + \left\|\boldsymbol{S} - \boldsymbol{U} + \frac{\boldsymbol{Y}_2}{\mu}\right\|_F^2\right). \end{aligned} \tag{9}$$

In the optimization process, the current target variable is updated while keeping other variables fixed. This method continuously optimizes each variable alternately, ensuring that each iteration moves toward the global optimal solution. This problem can be divided into the following subproblems.

1) With $\boldsymbol{U}$ and $\boldsymbol{E}$ fixed, update $\boldsymbol{S}$

$$\begin{aligned} &\boldsymbol{S}_{k+1} \\ &= \arg\min_{\boldsymbol{S}} \|\boldsymbol{S}\|_* \\ &\quad + \langle \nabla_{Sf}\left(\boldsymbol{S}_k, \boldsymbol{U}_k, \boldsymbol{E}_k, \boldsymbol{Y}_{1,k}, \boldsymbol{Y}_{2,k}, \mu_k\right), \boldsymbol{S} - \boldsymbol{S}_k\rangle \\ &\quad + \frac{\eta_1\mu_k}{2}\|\boldsymbol{S} - \boldsymbol{S}_k\|_F^2 \\ &= \arg\min_{\boldsymbol{S}} \|\boldsymbol{S}\|_* \\ &\quad + \frac{\eta_1\mu_k}{2}\left\|\boldsymbol{S} - \boldsymbol{S}_k + \left[-\boldsymbol{D}^T\left(\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_k - \boldsymbol{E}_k + \frac{\boldsymbol{Y}_{1,k}}{\mu_k}\right)\right.\right. \\ &\quad \left.\left. + \left(\boldsymbol{S}_k - \boldsymbol{U}_k + \frac{\boldsymbol{Y}_{2,k}}{\mu_k}\right)\right]\Big/\eta_1\right\|_F^2. \end{aligned} \tag{10}$$

In this step, the quadratic term f is replaced by the first-order approximation from the previous iteration, combined with a proximal operator. Here, ∇_{Sf} is the derivative of f with respect to $\boldsymbol{S}$, and $\eta_1 = ||D||_F^2$.

2) With $\boldsymbol{S}$ and $\boldsymbol{E}$ fixed, update $\boldsymbol{U}$

$$\boldsymbol{U}_{k+1} = \arg\min_{\boldsymbol{U}} \beta\|\boldsymbol{U}\|_1 + \frac{\mu_k}{2}\left\|\boldsymbol{S}_{k+1} - \boldsymbol{U} + \frac{\boldsymbol{Y}_{2,k}}{\mu_k}\right\|_F^2. \tag{11}$$

3) With $\boldsymbol{S}$ and $\boldsymbol{U}$ fixed, update $\boldsymbol{E}$

$$\begin{aligned} \boldsymbol{E}_{k+1} &= \arg\min_{\boldsymbol{E}} \lambda\|\boldsymbol{E}\|_{2,1} \\ &\quad + \frac{\mu_k}{2}\left\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_{k+1} - \boldsymbol{E} + \frac{\boldsymbol{Y}_{1,k}}{\mu_k}\right\|_F^2. \end{aligned} \tag{12}$$

4) Update Lagrange multipliers

$$\begin{aligned} \boldsymbol{Y}_{1,k+1} &= \boldsymbol{Y}_{1,k} + \mu_k(\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_{k+1} - \boldsymbol{E}_{k+1}) \\ \boldsymbol{Y}_{2,k+1} &= \boldsymbol{Y}_{2,k} + \mu_k(\boldsymbol{S}_{k+1} - \boldsymbol{U}_{k+1}). \end{aligned} \tag{13}$$

5) Update μ

$$\begin{aligned} &\mu_{k+1} \\ &= \min(\mu_{\max}, \rho\mu_k) \quad \text{where} \\ &\rho \\ &= \begin{cases} \rho_0, & \text{if } \mu_k \max(\sqrt{\eta_1}\|\boldsymbol{S}_{k+1} - \boldsymbol{S}_k\|_F, \|\boldsymbol{U}_{k+1} - \boldsymbol{U}_k\|_F, \\ & \qquad \|\boldsymbol{E}_{k+1} - \boldsymbol{E}_k\|_F)/\|\boldsymbol{X}\|_F \le \varepsilon_2 \\ 1, & \text{otherwise.} \end{cases} \end{aligned} \tag{14}$$

Although the LADMAP algorithm reduces the overall computational complexity by linearizing the quadratic penalty term, the LRASR still suffers from complex computation, and Nesterov acceleration is an excellent choice to speed up the LADMAP algorithm. The traditional Nesterov algorithm improves optimization efficiency and convergence by introducing a momentum term, which anticipates the gradient direction in each iteration to make precise adjustments. However, a fixed update magnitude cannot effectively accelerate the LADMAP algorithm. Therefore, we employ an adaptive update strategy to dynamically adjust the value of momentum, enhancing the performance of Nesterov. This adaptive strategy aims to better control the convergence speed during the iterative process by adaptively adjusting momentum and optimizing it in each iteration. The specific update rule is

$$\zeta_{k+1} = \begin{cases} \rho\zeta_k, & \text{if } \zeta_k < \zeta_{\max}, \quad \rho = \rho_0 \\ \zeta_{\max}, & \text{otherwise} \end{cases} \tag{15}$$

where ζ is the step size, $\zeta_{\max}$ is the upper bound of ζ, and the value of ρ is determined by the convergence conditions. The specific implementation of the algorithm is shown in Algorithm 1.

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Algorithm 1 ALADMAP

Require: Hyperspectral data $\boldsymbol{X}$, parameter $\beta > 0$, $\lambda > 0$

Initialization: $\boldsymbol{S}_0 = \boldsymbol{U}_0 = \boldsymbol{E}_0 = \boldsymbol{Y}_{1,0} = \boldsymbol{Y}_{2,0} = \boldsymbol{Z}_{1,0} = \boldsymbol{Z}_{2,0} = 0$, $\mu = 0.01$, $\mu_{\max} = 10^{10}$, $\rho_0 = 1.1$, $\epsilon_1 = 10^{-6}$, $\epsilon_2 = 10^{-2}$, $\eta_1 = \|\boldsymbol{D}\|_2^2$, $k = 0, \alpha_0 = 1, \zeta_0 = 1, \zeta_{max} = 2$

while $\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_0 - \boldsymbol{E}_0\|_F / \|\boldsymbol{X}\|_F \geq \epsilon_1$ **or** $\mu_k \max\left(\sqrt{\eta_1}\|\boldsymbol{S}_k - \boldsymbol{S}_{k-1}\|_F, \|\boldsymbol{U}_k - \boldsymbol{U}_{k-1}\|_F, \|\boldsymbol{E}_k - \boldsymbol{E}_{k-1}\|_F\right)/\|\boldsymbol{X}\|_F \geq \epsilon_2$ **do**

  Update $\boldsymbol{S}_{k+1}$

  $\boldsymbol{S}_{k+1} = \Theta_{(\eta_1\mu_k)^{-1}}(\boldsymbol{S}_k + [\boldsymbol{D}^T(\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_k - \boldsymbol{E}_k + \boldsymbol{Z}_{1,k}/\mu_k) - \left(\boldsymbol{S}_k - \boldsymbol{U}_k + \boldsymbol{Z}_{2,k}/\mu_k\right)]/\eta_1)$

  Update $\boldsymbol{U}_{k+1}$

  $\boldsymbol{U}_{k+1} = \mathcal{S}_{\beta\mu_k^{-1}}\left(\boldsymbol{S}_{k+1} + \boldsymbol{Z}_{2,k}/\mu_k\right)$

  Update $\boldsymbol{E}_{k+1}$

  $\boldsymbol{E}_{k+1} = \arg\min_{\boldsymbol{E}}(\lambda/\mu_k\|\boldsymbol{E}\|_{2,1} + \frac{1}{2}\|\boldsymbol{E} - (\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_{k+1} + \boldsymbol{Z}_{1,k}/\mu_k)\|_F^2$

  Update α_{k+1}

  $\alpha_{k+1} = (1 + \sqrt{1 + 4\alpha_k^2})/2$

  Update the Lagrangian multipliers

  $\boldsymbol{Y}_{1,k+1} = \boldsymbol{Z}_{1,k} + \mu_k(\boldsymbol{X} - \boldsymbol{D}\boldsymbol{S}_{k+1} - \boldsymbol{E}_{k+1})$

  $\boldsymbol{Y}_{2,k+1} = \boldsymbol{Z}_{2,k} + \mu_k(\boldsymbol{S}_{k+1} - \boldsymbol{U}_{k+1})$

  Update the momentum

  $\boldsymbol{Z}_{1,k+1} = \boldsymbol{Y}_{1,k+1} + \zeta_k(\alpha_{k-1}/\alpha_{k+1})(\boldsymbol{Y}_{1,k+1} - \boldsymbol{Y}_{1,k})$

  $\boldsymbol{Z}_{2,k+1} = \boldsymbol{Y}_{2,k+1} + \zeta_k(\alpha_{k-1}/\alpha_{k+1})(\boldsymbol{Y}_{2,k+1} - \boldsymbol{Y}_{2,k})$

  Update μ and ζ by equation (14) and (15)

end while

Ensure: $\boldsymbol{S}_k, \boldsymbol{U}_k, \boldsymbol{E}_k$

B. CPU-FPGA Optimized Implementation

FPGAs are flexible and configurable hardware platforms with excellent parallel processing capabilities and low-latency characteristics. They are suitable for applications with high parallel computing demands, such as digital signal processing, image processing, and deep learning, and outperform CPUs and GPUs in terms of energy efficiency and real-time processing. Compared to traditional CPU-based implementations, the CPU-FPGA hybrid platform provides a better balance between computational efficiency and flexibility. While CPUs offer strong general-purpose computing capabilities, they struggle with the highly iterative and matrix-intensive computations required for HAD. The FPGA, on the other hand, can be optimized for such workloads by leveraging customized parallel processing pipelines, significantly reducing computation time.

In contrast to GPU-based implementations, which are widely used for accelerating deep learning and image processing tasks, FPGAs exhibit superior energy efficiency and lower latency for irregular iterative computations. GPUs rely on massively parallel architectures with high memory bandwidth but may experience inefficiencies due to memory access bottlenecks and thread synchronization overhead in iterative optimization algorithms such as LADMAP. The FPGA's fine-grained parallelism and reconfigurable logic allow for dedicated hardware optimization, reducing redundant memory accesses and improving computational throughput.

In this article, we employ high-level synthesis (HLS) methods for FPGA development, utilizing high-level languages such as C/C++ to simplify the design process and accelerate the transition from concept to hardware implementation. Specifically, when using the k-means algorithm for clustering, multiple iterations are required until the final cluster centers converge. Therefore, we optimized the k-means computation process on the CPUs, hyperspectral data are partitioned for parallel computation, and the cluster centers are updated collaboratively. The specific implementation steps are given as follows.

1) In the initialization of cluster centers, k-means clustering generally selects k pixels randomly from $\boldsymbol{N}$ pixels to construct the initial cluster centers.
2) Each node computes the cluster assignment of pixels in its current partition in parallel. Specifically, the Euclidean distance between each pixel and the k cluster centers is calculated, and the pixel is assigned to the cluster with the nearest center.
3) Aggregate the data from each cluster, and update the cluster centers.
4) Repeat steps 2 and 3 until convergence conditions are met.

The overall architecture of the algorithm is shown in Fig. 2, with its implementation on FPGA focusing on data flow and resource utilization. The algorithm is based on a CPU-FPGA joint computing architecture, where the CPU handles relatively lightweight tasks, such as dictionary construction, while the FPGA processes the more computationally intensive LADMAP algorithm. In LRASR, LADMAP typically requires dozens to hundreds of iterations to converge, and utilizing the parallel computing capability of FPGA can significantly improve the processing efficiency of spectral data. To further enhance the performance of the algorithm on FPGA, we used Vivado HLS optimization directives, focusing on applying ARRAY PARTITION, PIPELINE, and UNROLL directives to enhance parallelism and data processing efficiency.

In the implementation of the algorithm, HSI is divided into central points and edge points that are processed independently. For large-scale datasets, the data are further partitioned into smaller blocks and processed in a pipelined manner, with each block independently updating variables such as $\boldsymbol{S}$. Double-buffering techniques are employed to overlap computation and data transfer, reducing memory access latency. The ARRAY PARTITION directive is used to split matrices such as $\boldsymbol{S}$ and $\boldsymbol{D}$ into subarrays, enabling parallel access and improving memory bandwidth utilization. Intermediate variables are stored in registers or on-chip BRAM to minimize interactions with external DDR, reducing power consumption and latency. Dynamic memory allocation is avoided by fixing buffer sizes in the FPGA logic, ensuring real-time performance.

Tasks are efficiently distributed across CPU and FPGA resources. During the dictionary construction phase, the k-means clustering task is parallelized by partitioning the data into subsets, each processed by a separate CPU thread. The operation of data partitioning can be performed in parallel on the CPU segment. On the FPGA side, hardware parallelism is maximized through pipelining and data-level parallelism. Matrix operations in LADMAP iterations are decomposed into multiple pipeline stages, with each stage processing

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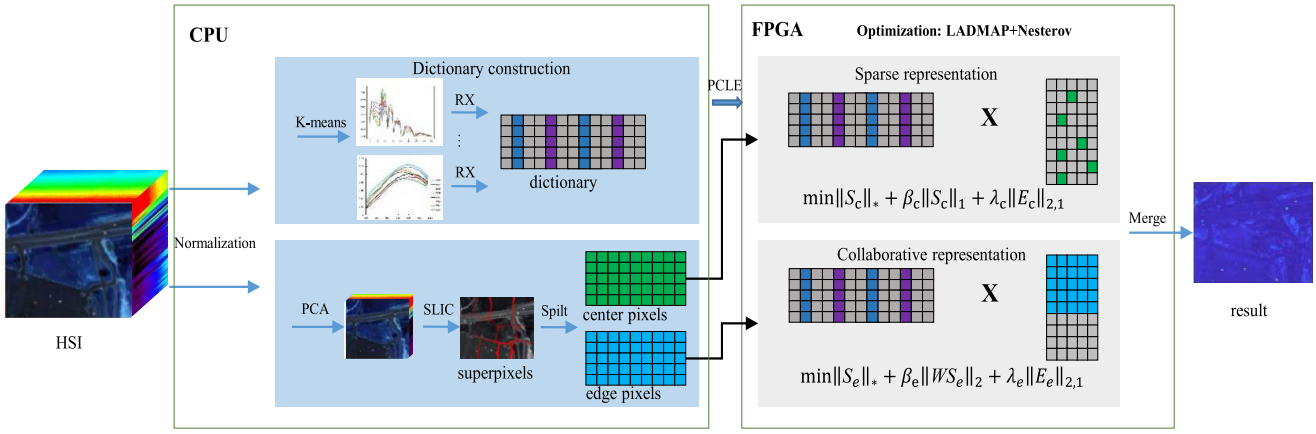
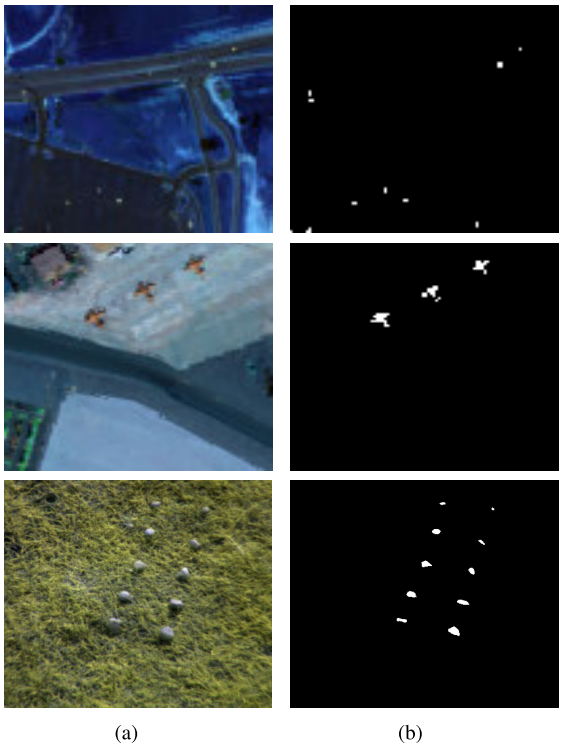


Fig. 2. Overall algorithm architecture diagram.



(a) (b)

Fig. 3. Datasets and ground truth: (a) image scene and (b) ground truth.

different data blocks to achieve instruction-level parallelism. Low-dimensional operations, such as vector dot products, are unrolled using the UNROLL directive, enabling multiple PEs to process different data elements simultaneously. Task-level parallelism is also exploited by assigning the model-solving tasks for central points and edge points to independent computing units on the FPGA, allowing them to execute concurrently.

FPGAs also have certain limitations, particularly their small memory size, which makes them unsuitable for processing large amounts of hyperspectral data simultaneously. In the representation framework proposed in this article, the original image data are divided into central points and edge points, which can be computed independently. This significantly reduces the memory requirement compared to processing the entire image. In cases where the memory cannot accommodate the data, a blockwise processing approach could be used to accelerate the process, and each block of data is processed in a pipelined manner, with each block independently updating the variables $\boldsymbol{S}, \boldsymbol{U}$, and $\boldsymbol{E}$, as well as auxiliary variables. Liu et al. [32] also demonstrated that the parallel LRASR does not reduce the accuracy of the algorithm.

TABLE I
IMAGE ATTRIBUTES USED FOR THE EXPERIMENTS

Images	Sensor	Size	Bands	SR[m]
Urban	HYDICE	80×100	210	2
San Diego	AVIRIS	100×100	224	3.5
Cri	AVIRIS	400×400	46	3.4

V. EXPERIMENTS

In this section, we experimentally verify the performance of the proposed method on real hyperspectral datasets. The main content includes a detailed introduction of the datasets, an evaluation of the method on these datasets, and a comparison of the runtime after parallel optimization. The experimental platform used an Intel Core i7-9700K @ 3.6-GHz processor and an FPGA from the Xilinx Zynq-7000 development board (model AX7Z035B) to demonstrate the efficiency of the method on different hardware platforms.

A. Datasets

Table I provides detailed information on the key characteristics of different datasets, including the sensor type used, size, bands, and spectral resolution. Fig. 3 shows the three datasets along with their corresponding ground truth.

The Urban dataset primarily comprises materials such as buildings, grass, roads, and trees; this dataset was acquired by the HYDICE airborne sensor, which is mounted on an aircraft platform and specialized in hyperspectral digital imagery collection. For the experiments, an image with spatial dimensions of 80 × 100 pixels was utilized. Prior to analysis, certain bands affected by water vapor noise (specifically bands 1–4, 76, 87, 101–111, 136–153, and 198–210) were removed, leaving a

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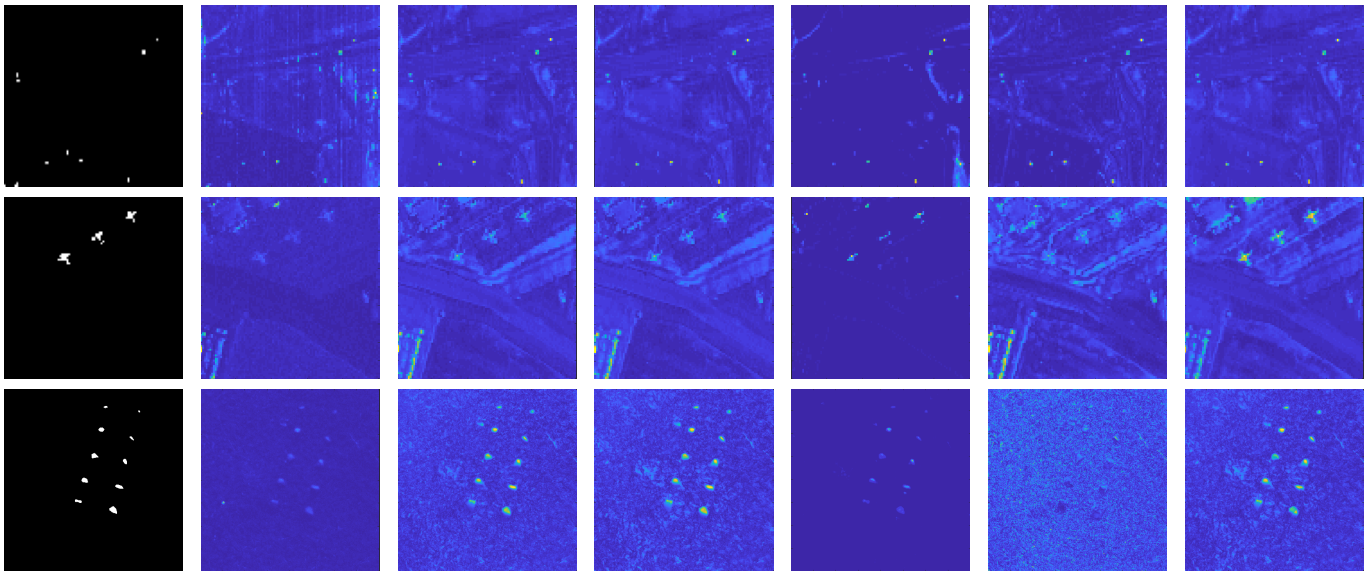


Fig. 4. HAD predictions in experimental images. (a) Ground truth. (b) RX. (c) LRCRD. (d) LRASR. (e) IEEPST. (f) GTVLRR. (g) Proposed.

total of 162 bands for further examination. Within this refined dataset, cars are identified as anomalous targets, with a total count of 21 such instances.

The San Diego dataset, captured by the AVIRIS sensor, features a spatial resolution of 3.5 m and spans wavelengths from 370 to 2510 nm. The experimental area measures 100 × 100 pixels across 186 spectral bands, with water absorption areas and defective bands meticulously removed. This dataset includes diverse scenes such as rooftops of various shapes, different types of tarmacs, an airport runway, and some vegetation. In this dataset, airplanes are considered anomalous targets.

The Cri was collected by the Nuance Cri hyperspectral sensor with a spectral resolution of 10 nm and wavelengths ranging from 650 to 1100 nm. This dataset has a spatial size of 400 × 400 pixels and includes 46 spectral bands. The Cri dataset primarily consists of two materials: grass and rocks, with the rocks identified as the anomalous targets. Within this dataset, there are 2216 pixels classified as anomalies.

B. Experimental Settings

In the comparative experiments, we selected several representative and advanced hyperspectral detection algorithms, which are based on representation learning and deep learning, to compare with the proposed method. The selected comparative algorithms include RX [4], LRCRD [33], LRASR [15], IEEPST [34], and GTVLRR [35] detectors. For the evaluation of detection performance, we quantitatively assess the detection capability using the area under the curve (AUC) value, presenting the detection result plots and AUC values.

C. Accuracy Evaluation

The experiments were conducted on different datasets, and Fig. 4 shows the detection results for three datasets, with the results for the Urban, San Diego, and Cri datasets presented from top to bottom. The leftmost column of the figure shows the ground truth maps of the datasets, the second column shows the detection results of the RX algorithm, the third column shows the LRCRD results, the fourth column shows the LRASR results, the fifth column shows the IEEPST results, the sixth column shows the GTVLRR results, and the last column represents the detection results of the proposed method. It can be observed that the detection results of the proposed method are the clearest and demonstrate the best performance.

For the Urban dataset, the RX algorithm has average detection performance, with indistinct separation between the background and anomalies. LRCRD and LRASR are able to detect the anomalies generally, but there are some issues with individual anomalies being less distinct. IEEPST achieves the best background suppression, but the anomaly scores are not high enough, making the results less evident. GTVLRR performs well in detection, but some background points have relatively high anomaly coefficients. The proposed method effectively suppresses the background, with a low anomaly target miss rate.

For the San Diego dataset, the RX detected targets are not clear enough, and the background is relatively prominent in the LRCRD and LRASR results, with the anomalies not standing out sufficiently. IEEPST can suppress the background, but the targets are not displayed completely, lacking discernibility. GTVLRR does not clearly distinguish between the background and targets, making them easy to confuse. The proposed algorithm highlights the anomalous target information; although there are points in the background with high anomaly residuals, the anomalies are more distinct.

For the Cri dataset, which consists of only two types of materials, the RX algorithm effectively distinguishes background points. However, the edge pixels of the anomalies exhibit significant mixing with the background, leading to ambiguity. LRCRD and LRASR show clear anomalies, and the

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TABLE II
AUC VALUES OF THE COMPARABLE METHODS FOR THREE DATASETS

Images	RX	LRCRD	LRASR	IEEPST	GTVLRR	Proposed
Urban	0.8586	0.9186	0.9209	0.9606	0.9455	0.9771
San Diego	0.9759	0.9943	0.9946	0.9463	0.9746	0.9949
Cri	0.9135	0.9911	0.9901	0.9991	0.6191	0.9998

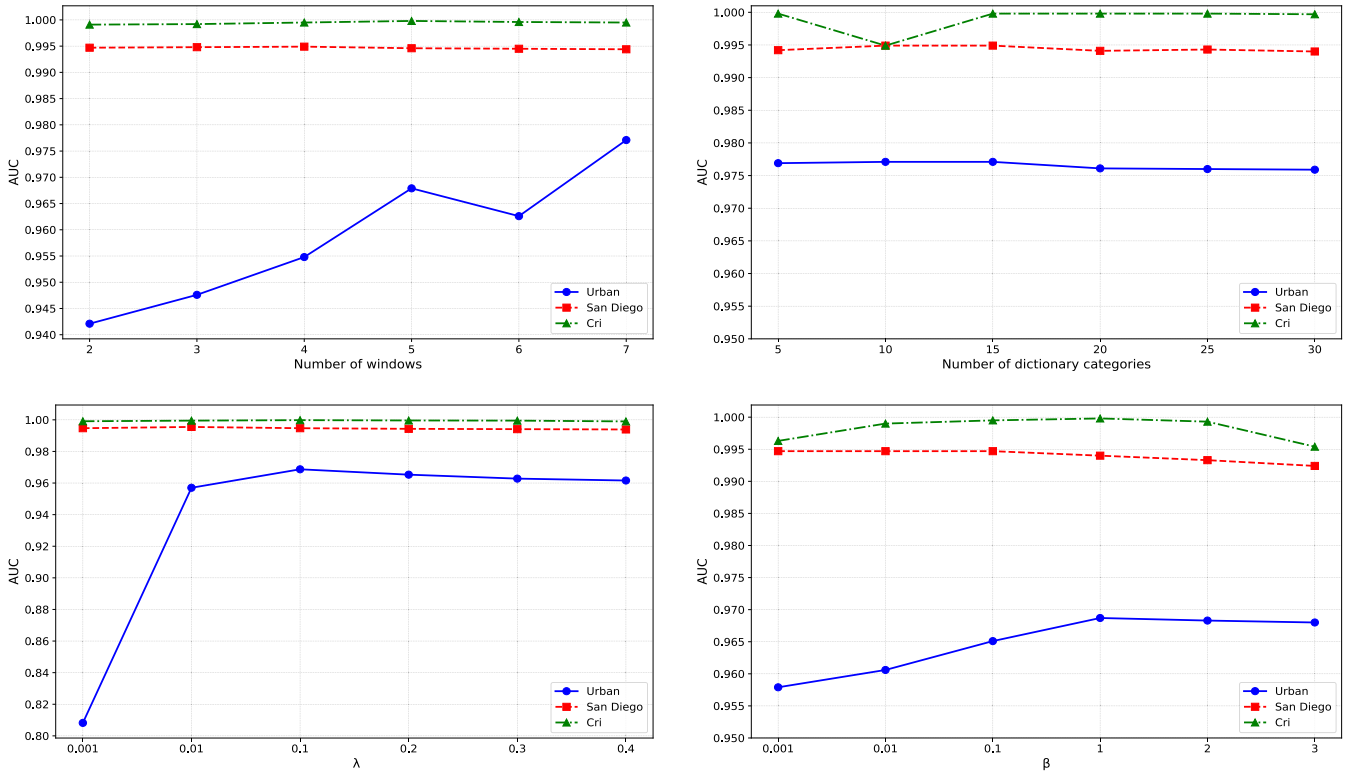


Fig. 5. Results of the parametric analysis.

background and targets can generally be separated. IEEPST performs similar to RX, with unclear targets. GTVLRR performs poorly in relatively simple datasets, with low detection performance and difficulty distinguishing between background and anomalies. The proposed method effectively displays the anomalous target pixels, with clearer handling of edge pixels.

To more accurately evaluate the detection performance of the proposed method, we calculated the AUC values of each algorithm using the anomaly detection images. The results are presented in Table II. From the results in the table, it is clear that the proposed method achieved the best results across all detection datasets. On the Urban dataset, compared to the baseline algorithms LRASR and LRCRD, the proposed method achieved a significant accuracy improvement. Compared to IEEPST, which had similar accuracy, the proposed method is clearer and easier to distinguish. On the other two datasets, the proposed algorithm also achieved very good detection accuracy. This performance improvement can be attributed to the pixel-segmentation-based representation framework used in this work. Compared to traditional low-rank methods, this approach considers the different representation capabilities of the dictionary for various pixels, thereby enhancing the representation ability for HSIs and improving the AUC values.

D. Parametric Analysis

Fig. 5 presents the results of the parametric analysis, illustrating the influence of four key hyperparameters—the number of windows, the number of dictionary categories, λ, and β. Most of them are robust to the detection results, so we discuss them collectively without distinguishing among the models.

For the number of windows, San Diego and Cri datasets maintain a consistently high AUC, implying that they are less sensitive to this parameter. Minor fluctuations at certain points indicate that an excessively large window size may not always be beneficial. However, in the Urban dataset, larger window sizes yield better results. This indicates that the mixed pixels in this dataset are more complex, suggesting that the initial dictionary was not sufficiently comprehensive.

For the number of dictionary categories, the AUC for the Urban dataset remains relatively stable across different dictionary sizes, suggesting that increasing the number of dictionary categories does not significantly impact the detection performance. For Cri, slight variations are observed, but, overall, the performance remains robust. Similarly, San Diego exhibits a nearly flat trend, indicating that the dictionary complexity has a minimal effect on these datasets.

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TABLE III
AVERAGE COMPUTING TIME (IN SECONDS) OF THE COMPARED METHODS

Images	RX	LRCRD	LRASR	IEEPST	GTVLRR	Proposed
Urban	0.54	91.46	88.46	46.77	120.99	41.64
San Diego	0.48	89.40	72.64	49.44	116.61	45.01
Cri	0.70	265.31	243.43	47.39	2500.98	651.22

For λ, it plays a critical role in balancing sparsity and reconstruction accuracy. For the Urban dataset, a small λ (0.001) results in poor performance, but, as it increases, the AUC improves significantly and stabilizes beyond 0.01. This trend indicates that an appropriate level of regularization enhances model generalization. Meanwhile, the San Diego and Cri datasets show minimal variations, suggesting that they are less influenced by λ.

For β, it exhibits a positive impact on the Urban dataset, with AUC increasing as β grows, followed by a slight saturation beyond $\beta = 1$. This suggests that weighting adjustments contribute to improved detection but have diminishing returns at higher values. The Cri dataset shows minor fluctuations, while San Diego remains largely stable, reflecting its robustness to β variations.

E. Performance Evaluation

In this section, we delve into the assessment of the processing efficiency of the algorithm, focusing specifically on the evaluation of the ALADMAP algorithm. The primary objective is to compute the efficiency of each algorithm. As shown in Table III, due to the segmentation of the image into two parts, the efficiency of our method is notably high. While the RX algorithm achieves the shortest execution time, its detection accuracy is often compromised, limiting its practical applicability. For the Cri dataset, the execution time increases significantly compared to other datasets, as its larger spatial dimensions impose higher computational demands. In addition, low-rank representation-based methods typically rely on iterative optimization, where the convergence error directly impacts the execution time. The experimental results presented in this study are based on a unified convergence criterion, which may not highlight the time efficiency of low-rank methods compared to RX. However, in practical applications, the convergence threshold can be adjusted to significantly reduce computation time. For instance, by relaxing the convergence criteria, the execution time can be accelerated by five to ten times, with a negligible accuracy loss of less than 1%. This flexibility underscores the practical value of the proposed method, as it balances computational efficiency and detection performance effectively.

Then, we compare the performance of ALADMAP against the standard LADMAP optimization algorithm across various datasets. Fig. 6 presents a comprehensive visualization of the convergence curves for both ALADMAP and LADMAP on different datasets. Upon examining these curves, it becomes apparent that the proposed ALADMAP algorithm demonstrates superior performance compared to LADMAP across all evaluated datasets. Notably, ALADMAP reduces the original convergence process from approximately 100 iterations to around 80 iterations, marking a significant 20% improvement in performance.

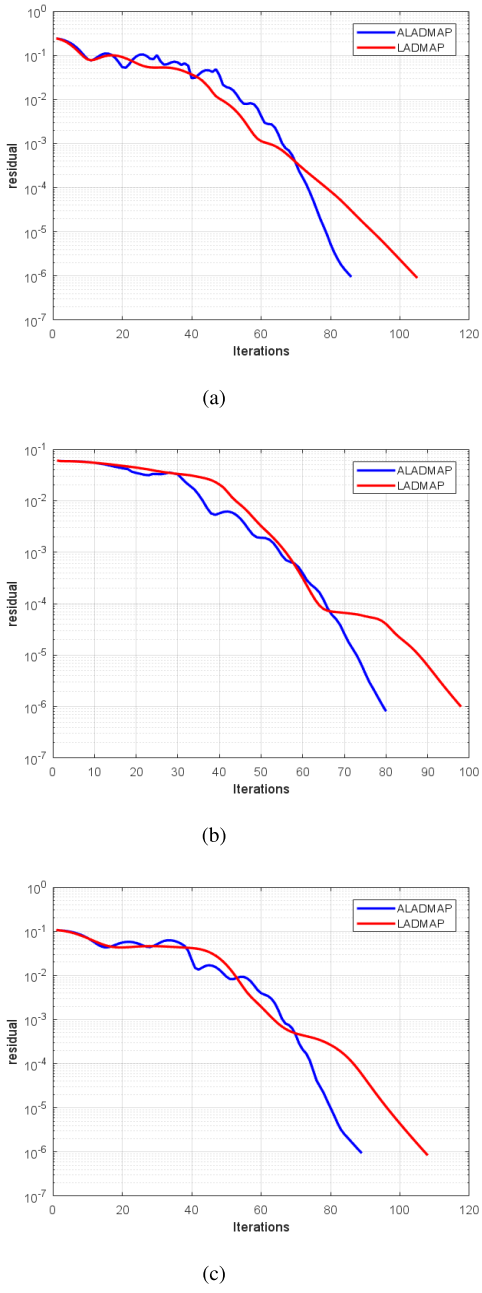


Fig. 6. Convergence curves for ALADMAP and LADMAP methods. (a) Urban. (b) San Diego. (c) Cri.

A closer inspection of the convergence curves reveals intriguing insights. For both datasets, during the initial 65

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TABLE IV
EXECUTION TIME (IN SECONDS) COMPARISON

Dataset	MATLAB	C++	GPU	CPU-FPGA
Urban	41.64	1647.26	21.64	13.37
San Diego	45.01	2313.16	25.17	18.10
Cri	651.22	9948.52	262.34	189.63

iterations, both methods exhibit comparable performance, with their convergence curves closely tracking each other. However, ALADMAP begins to show a distinct advantage earlier in the iteration process and maintains a consistently faster downward trend after reaching 70 iterations. This suggests that ALADMAP not only accelerates the convergence rate but also achieves a lower residual value more swiftly, which is crucial for enhancing both the efficiency and accuracy of the proposed algorithm. The adaptive step size Nesterov technique integrated into ALADMAP allows for more dynamic and efficient adjustments during the optimization process, enabling the algorithm to navigate the loss landscape more effectively. This results in faster convergence and improved performance, as evidenced by the lower residual values achieved by ALADMAP compared to LADMAP.

Table IV presents the execution times of the proposed algorithm across different processing platforms and programming environments. The CPU-based implementation was conducted using MATLAB and C++, while the GPU-accelerated implementation was executed in Python on an NVIDIA 2080Ti. These data allow for a thorough analysis of the algorithm's time efficiency and adaptability across different hardware platforms, providing valuable practical insights for HAD. In the MATLAB environment, the processing times were 41.64 s for the Urban dataset, 45.01 s for the San Diego dataset, and 651.22 s for the larger Cri dataset. The significantly longer processing time for the Cri dataset is primarily due to its larger data size and greater spatial dimensions compared to the Urban and San Diego datasets. During SVD computation, the most time-consuming part is the matrix decomposition, leading to a significant increase in computation time. When the algorithm was implemented in a C++ environment, the processing time increased significantly: 1647.26 s for the Urban dataset, 2313.16 s for the San Diego dataset, and 9948.52 s for the Cri dataset. This difference may be partially attributed to MATLAB's optimization mechanisms, which the C++ environment may lack. GPU acceleration can significantly reduce the execution time of the algorithm, with processing times for the three datasets being 21.64 s for the Urban dataset, 25.17 s for the San Diego dataset, and 262.34 s for the Cri dataset. However, the overall execution efficiency is still lower than the CPU-FPGA implementation. Deploying the algorithm on the CPU-FPGA architecture improved its performance. In the CPU-FPGA joint computing mode, the processing time for the Urban dataset was reduced to 13.37 s, for the San Diego dataset to 18.10 s, and for the Cri dataset to just 189.63 s. This demonstrates that FPGA, as parallel computing hardware, can greatly enhance code execution efficiency, especially for computationally intensive tasks such as HAD, providing an efficient solution for real-time analysis. However, it is worth noting that despite the significant advantages of FPGA in terms of time efficiency, it may be limited by memory and computational resources when processing large-scale or high-band HSI. When balancing algorithm accuracy and time efficiency, the proposed algorithm excels in high-accuracy detection. Although it may not match the time efficiency of some real-time detection algorithms, it has a significant advantage in terms of detection accuracy.

VI. CONCLUSION

This article proposes and validates an efficient low-rank representation HAD based on pixel segmentation, aimed at achieving low-rank model construction for detection tasks with high accuracy and efficiency. By segmenting different pixels to account for the varying sensitivities of the dictionary to different pixels, better detection results are obtained. Building on this, we further introduced the adaptive step size accelerated ALADMAP algorithm, significantly enhancing the convergence speed of the optimization. Furthermore, the algorithm was deployed on a CPU-FPGA joint computing platform, leveraging the parallel processing and high-speed computation capabilities of the FPGA to accelerate image processing, providing a feasible and efficient solution for low-rank modeling and hyperspectral data processing.

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Kun Yu was born in Jiangsu, China, in 1995. He received the B.Sc. degree in computer science and technology from the School of Computer Science and Technology, Huaiyin Normal University, Huaian, China, in 2018. He is currently pursuing the Ph.D. degree with Nanjing University of Science and Technology (NJUST), Nanjing, China.

His research interests include hyperspectral image processing, parallel computing, and big data processing.



Zebin Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science and technology from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 2003 and 2007, respectively.

He was a Visiting Scholar with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, Cáceres, Spain, from June 2014 to 2015; the Department of Mathematics, University of California at Los Angeles, Los Angeles, CA, USA, from August 2016 to September 2016 and from July 2017 to August 2017; and the GIPSA-Lab, Grenoble INP, Université Grenoble Alpes, Grenoble, France, from August 2018 to September 2018. He is currently a Professor with the School of Computer Science and Engineering, NJUST. His research interests include hyperspectral image processing, parallel computing, big data processing, and their applications in railway foreign object detection.



Jin Sun (Member, IEEE) received the B.S. and M.S. degrees in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 2004 and 2006, respectively, and the Ph.D. degree in electrical and computer engineering from The University of Arizona, Tucson, AZ, USA, in 2011.

From January 2012 to September 2014, he was a Technical Staff Member with Orora Design Technologies Inc., Redmond, WA, USA. He is currently a Professor with the School of Computer Science and Engineering, NJUST. His research interests include cloud computing, edge computing and edge intelligence, and embedded systems.



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently a Lecturer with the School of Computer Science and Engineering, NJUST. His research interests include hyperspectral image (HSI) classification, hyperspectral detection, image processing, machine learning, and their applications in railway foreign object detection.

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Yi Zhang received the B.S. and Ph.D. degrees from the School of Computer Science and Engineering, Southeast University, Nanjing, China, in 2005 and 2011, respectively.

In 2009, he was an Intern with the IBM China Research Laboratory, Beijing, China. In 2011, he joined Huawei Tech. Company, Nanjing, as a Technical Research Staff Member. He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing. His research interests include project scheduling, workflow optimization, and resource management and allocation in cloud computing, mobile computing, and edge computing.



Zhihui Wei (Member, IEEE) was born in Jiangsu, China, in 1963. He received the B.Sc. and M.Sc. degrees in applied mathematics and the Ph.D. degree in communication and information systems from Southeast University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently a Professor and a Doctoral Supervisor with Nanjing University of Science and Technology (NJUST), Nanjing. His research interests include partial differential equations, mathematical image processing, multiscale analysis, sparse representation, compressive sensing, and their applications in railway foreign object detection.



Peng Zheng was born in Zhejiang, China, in 1994. He received the B.Sc. degree in computer science and technology from the College of Information Management, Nanjing Agricultural University, Nanjing, China, in 2016. He is currently pursuing the Ph.D. degree with Nanjing University of Science and Technology (NJUST), Nanjing.

His research interests are tensor decomposition and distributed computing and fusion.

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